



## **FATE MONTHLY REPORT**

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# **FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE**

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Performance assessment  
Annex  
November 2025



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## 1 NIGHT1: Contingency tables

Tables in section 1 are included in the range: 1-22.

| wind speed<br>$ms^{-1}$                                                         |                | Observations |                |          |
|---------------------------------------------------------------------------------|----------------|--------------|----------------|----------|
|                                                                                 |                | ws < 4.2     | 4.2 < ws < 7.6 | ws > 7.6 |
| Model data                                                                      | ws < 4.2       | 7445         | 2793           | 414      |
|                                                                                 | 4.2 < ws < 7.6 | 4700         | 6160           | 2008     |
|                                                                                 | ws > 7.6       | 511          | 2571           | 9828     |
| Number of points 36430 ; PC=64.3%; EBD=2.5%; POD1=58.8%; POD2=53.5%; POD3=80.2% |                |              |                |          |

Table 1: night1 - Contingency table for wind speed ( $ms^{-1}$ ) in standard configuration

| wind speed                                                                      |                | Observations |                |          |
|---------------------------------------------------------------------------------|----------------|--------------|----------------|----------|
| $ms^{-1}$                                                                       |                | ws < 4.2     | 4.2 < ws < 7.6 | ws > 7.6 |
| Model data                                                                      | ws < 4.2       | 11777        | 827            | 0        |
|                                                                                 | 4.2 < ws < 7.6 | 879          | 10116          | 638      |
|                                                                                 | ws > 7.6       | 0            | 581            | 11612    |
| Number of points 36430 ; PC=92.0%; EBD=0.0%; POD1=93.1%; POD2=87.8%; POD3=94.8% |                |              |                |          |

Table 2: night1 - Contingency table for wind speed ( $ms^{-1}$ ) processed with AR (1H)

| wind speed<br>$ms^{-1}$                                                         | Observations     |                  |           |     |
|---------------------------------------------------------------------------------|------------------|------------------|-----------|-----|
|                                                                                 | ws < 12.0        | 12.0 < ws < 18.0 | ws > 18.0 |     |
|                                                                                 | ws < 12.0        | 32201            | 325       | 0   |
| Model data                                                                      | 12.0 < ws < 18.0 | 290              | 3040      | 75  |
|                                                                                 | ws > 18.0        | 0                | 76        | 423 |
| Number of points 36430 ; PC=97.9%; EBD=0.0%; POD1=99.1%; POD2=88.3%; POD3=84.9% |                  |                  |           |     |

Table 3: night1 - Contingency table for wind speed ( $ms^{-1}$ ) processed with AR (1H) and thresholds (12.0,18.0)

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| wind direction<br>degree                                                                    |       | North | East | South | Observations<br>West |
|---------------------------------------------------------------------------------------------|-------|-------|------|-------|----------------------|
| Model data                                                                                  | North | 16040 | 1063 | 25    | 222                  |
|                                                                                             | East  | 1535  | 2557 | 527   | 10                   |
|                                                                                             | South | 159   | 643  | 4107  | 318                  |
|                                                                                             | West  | 561   | 142  | 45    | 314                  |
| Number of points 28268 ; PC=81.4%; EBD=1.2%; POD1=87.7%; POD2=58.0%; POD3=87.3%; POD4=36.3% |       |       |      |       |                      |

Table 4: night1 - Contingency table for wind direction (degree) in standard configuration

| wind direction<br>degree                                                                    |       | North | East | South | Observations<br>West |
|---------------------------------------------------------------------------------------------|-------|-------|------|-------|----------------------|
| Model data                                                                                  | North | 17935 | 263  | 0     | 67                   |
|                                                                                             | East  | 327   | 4019 | 99    | 0                    |
|                                                                                             | South | 0     | 123  | 4582  | 24                   |
|                                                                                             | West  | 33    | 0    | 23    | 773                  |
| Number of points 28268 ; PC=96.6%; EBD=0.0%; POD1=98.0%; POD2=91.2%; POD3=97.4%; POD4=89.5% |       |       |      |       |                      |

Table 5: night1 - Contingency table for wind direction (degree) processed with AR (1H)

| relative humidity<br>percent                                                    |                 | Observations |                 |           |
|---------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                                 |                 | rh < 7.1     | 7.1 < rh < 15.3 | rh > 15.3 |
| Model data                                                                      | rh < 7.1        | 8630         | 2472            | 19        |
|                                                                                 | 7.1 < rh < 15.3 | 5485         | 7137            | 1326      |
|                                                                                 | rh > 15.3       | 667          | 3466            | 6852      |
| Number of points 36054 ; PC=62.7%; EBD=1.9%; POD1=58.4%; POD2=54.6%; POD3=83.6% |                 |              |                 |           |

Table 6: night1 - Contingency table for relative humidity (percent) in standard configuration

| relative humidity<br>percent                                                    |                 | Observations |                 |           |
|---------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                                 |                 | rh < 7.1     | 7.1 < rh < 15.3 | rh > 15.3 |
| Model data                                                                      | rh < 7.1        | 13996        | 594             | 4         |
|                                                                                 | 7.1 < rh < 15.3 | 784          | 12097           | 258       |
|                                                                                 | rh > 15.3       | 2            | 384             | 7935      |
| Number of points 36054 ; PC=94.4%; EBD=0.0%; POD1=94.7%; POD2=92.5%; POD3=96.8% |                 |              |                 |           |

Table 7: night1 - Contingency table for relative humidity (percent) processed with AR (1H)

| relative humidity<br>percent                                                    |                  | Observations |                  |           |
|---------------------------------------------------------------------------------|------------------|--------------|------------------|-----------|
|                                                                                 |                  | rh < 50.0    | 50.0 < rh < 70.0 | rh > 70.0 |
| Model data                                                                      | rh < 50.0        | 35397        | 51               | 0         |
|                                                                                 | 50.0 < rh < 70.0 | 54           | 483              | 9         |
|                                                                                 | rh > 70.0        | 0            | 7                | 53        |
| Number of points 36054 ; PC=99.7%; EBD=0.0%; POD1=99.8%; POD2=89.3%; POD3=85.5% |                  |              |                  |           |

Table 8: night1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

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| precipitable water vapor<br>mm                                                  | Observations    |                 |           |
|---------------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                                 | pwv < 1.9       | 1.9 < pwv < 3.5 | pwv > 3.5 |
| Model data                                                                      | pwv < 1.9       | 9279            | 1731      |
|                                                                                 | 1.9 < pwv < 3.5 | 1375            | 4426      |
|                                                                                 | pwv > 3.5       | 62              | 925       |
| Number of points 24755 ; PC=80.1%; EBD=0.3%; POD1=86.6%; POD2=62.5%; POD3=88.1% |                 |                 |           |

Table 9: night1 - Contingency table for precipitable water vapor (mm) in standard configuration

| precipitable water vapor<br>mm                                                  | Observations    |                 |           |
|---------------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                                 | pwv < 1.9       | 1.9 < pwv < 3.5 | pwv > 3.5 |
| Model data                                                                      | pwv < 1.9       | 10551           | 232       |
|                                                                                 | 1.9 < pwv < 3.5 | 165             | 6769      |
|                                                                                 | pwv > 3.5       | 0               | 81        |
| Number of points 24755 ; PC=97.7%; EBD=0.0%; POD1=98.5%; POD2=95.6%; POD3=98.6% |                 |                 |           |

Table 10: night1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

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| total seeing<br>arcsec                                                          | Observations |                 |           |
|---------------------------------------------------------------------------------|--------------|-----------------|-----------|
|                                                                                 | see < 0.6    | 0.6 < see < 0.8 | see > 0.8 |
| see < 0.6                                                                       | 4206         | 3492            | 1225      |
| Model data 0.6 < see < 0.8                                                      | 2853         | 3582            | 2986      |
| see > 0.8                                                                       | 474          | 1226            | 3108      |
| Number of points 23152 ; PC=47.1%; EBD=7.3%; POD1=55.8%; POD2=43.2%; POD3=42.5% |              |                 |           |

Table 11: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

| total seeing<br>arcsec                                                          | Observations |                 |           |
|---------------------------------------------------------------------------------|--------------|-----------------|-----------|
|                                                                                 | see < 0.6    | 0.6 < see < 0.8 | see > 0.8 |
| see < 0.6                                                                       | 7489         | 1234            | 39        |
| Model data 0.6 < see < 0.8                                                      | 1028         | 6295            | 736       |
| see > 0.8                                                                       | 10           | 828             | 6852      |
| Number of points 23152 ; PC=84.2%; EBD=0.2%; POD1=87.8%; POD2=75.3%; POD3=89.8% |              |                 |           |

Table 12: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.0

| total seeing<br>arcsec                                                          | Observations |                 |           |
|---------------------------------------------------------------------------------|--------------|-----------------|-----------|
|                                                                                 | see < 0.6    | 0.6 < see < 0.8 | see > 0.8 |
| see < 0.6                                                                       | 7169         | 3               | 855       |
| Model data 0.6 < see < 0.8                                                      | 0            | 8089            | 853       |
| see > 0.8                                                                       | 364          | 208             | 6464      |
| Number of points 23152 ; PC=93.8%; EBD=5.3%; POD1=95.2%; POD2=97.5%; POD3=79.1% |              |                 |           |

Table 13: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

| total seeing<br>arcsec                                                          | Observations |                 |           |
|---------------------------------------------------------------------------------|--------------|-----------------|-----------|
|                                                                                 | see < 0.6    | 0.6 < see < 0.8 | see > 0.8 |
| see < 0.6                                                                       | 8518         | 30              | 34        |
| Model data 0.6 < see < 0.8                                                      | 0            | 8304            | 33        |
| see > 0.8                                                                       | 9            | 23              | 7593      |
| Number of points 23152 ; PC=99.6%; EBD=0.2%; POD1=99.9%; POD2=99.4%; POD3=99.1% |              |                 |           |

Table 14: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.24

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| wavefront coherence time                                                        |                    | Observations |                    |              |
|---------------------------------------------------------------------------------|--------------------|--------------|--------------------|--------------|
| ms                                                                              |                    | $\tau < 3.4$ | $3.4 < \tau < 5.8$ | $\tau > 5.8$ |
| Model data                                                                      | $\tau < 3.4$       | 3617         | 2408               | 208          |
|                                                                                 | $3.4 < \tau < 5.8$ | 1391         | 4344               | 2417         |
|                                                                                 | $\tau > 5.8$       | 211          | 2003               | 5375         |
| Number of points 21974 ; PC=60.7%; EBD=1.9%; POD1=69.3%; POD2=49.6%; POD3=67.2% |                    |              |                    |              |

Table 15: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

| wavefront coherence time                                                        |                    | Observations |                    |              |
|---------------------------------------------------------------------------------|--------------------|--------------|--------------------|--------------|
| ms                                                                              |                    | $\tau < 3.4$ | $3.4 < \tau < 5.8$ | $\tau > 5.8$ |
| Model data                                                                      | $\tau < 3.4$       | 4807         | 442                | 5            |
|                                                                                 | $3.4 < \tau < 5.8$ | 412          | 7780               | 589          |
|                                                                                 | $\tau > 5.8$       | 0            | 533                | 7406         |
| Number of points 21974 ; PC=91.0%; EBD=0.0%; POD1=92.1%; POD2=88.9%; POD3=92.6% |                    |              |                    |              |

Table 16: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 0.0

| wavefront coherence time                                                        |                    | Observations |                    |              |
|---------------------------------------------------------------------------------|--------------------|--------------|--------------------|--------------|
| ms                                                                              |                    | $\tau < 3.4$ | $3.4 < \tau < 5.8$ | $\tau > 5.8$ |
| Model data                                                                      | $\tau < 3.4$       | 4671         | 285                | 945          |
|                                                                                 | $3.4 < \tau < 5.8$ | 337          | 7395               | 945          |
|                                                                                 | $\tau > 5.8$       | 211          | 1075               | 7055         |
| Number of points 21974 ; PC=87.0%; EBD=5.3%; POD1=89.5%; POD2=84.5%; POD3=78.9% |                    |              |                    |              |

Table 17: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

| wavefront coherence time                                                        |                    | Observations |                    |              |
|---------------------------------------------------------------------------------|--------------------|--------------|--------------------|--------------|
| ms                                                                              |                    | $\tau < 3.4$ | $3.4 < \tau < 5.8$ | $\tau > 5.8$ |
| Model data                                                                      | $\tau < 3.4$       | 5213         | 5                  | 43           |
|                                                                                 | $3.4 < \tau < 5.8$ | 6            | 8694               | 43           |
|                                                                                 | $\tau > 5.8$       | 0            | 56                 | 7957         |
| Number of points 21974 ; PC=99.5%; EBD=0.2%; POD1=99.9%; POD2=99.3%; POD3=98.9% |                    |              |                    |              |

Table 18: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 1.22

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| ground layer fraction<br>dimensionless                                           | Observations    |                 |           |
|----------------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                                  | glf < 0.6       | 0.6 < glf < 0.7 | glf > 0.7 |
| Model data                                                                       | glf < 0.6       | 3368            | 3843      |
|                                                                                  | 0.6 < glf < 0.7 | 2202            | 2922      |
|                                                                                  | glf > 0.7       | 489             | 1091      |
| Number of points 22195 ; PC=36.3%; EBD=15.3%; POD1=55.6%; POD2=37.2%; POD3=21.4% |                 |                 |           |

Table 19: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

| ground layer fraction<br>dimensionless                                          | Observations    |                 |           |
|---------------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                                 | glf < 0.6       | 0.6 < glf < 0.7 | glf > 0.7 |
| Model data                                                                      | glf < 0.6       | 5140            | 1149      |
|                                                                                 | 0.6 < glf < 0.7 | 871             | 5585      |
|                                                                                 | glf > 0.7       | 48              | 1122      |
| Number of points 22195 ; PC=79.4%; EBD=0.4%; POD1=84.8%; POD2=71.1%; POD3=83.3% |                 |                 |           |

Table 20: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

| ground layer fraction<br>dimensionless                                           | Observations    |                 |           |
|----------------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                                  | glf < 0.6       | 0.6 < glf < 0.7 | glf > 0.7 |
| Model data                                                                       | glf < 0.6       | 5501            | 1077      |
|                                                                                  | 0.6 < glf < 0.7 | 69              | 6744      |
|                                                                                  | glf > 0.7       | 489             | 35        |
| Number of points 22195 ; PC=78.3%; EBD=16.3%; POD1=90.8%; POD2=85.8%; POD3=47.7% |                 |                 |           |

Table 21: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

| ground layer fraction<br>dimensionless                                          | Observations    |                 |           |
|---------------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                                 | glf < 0.6       | 0.6 < glf < 0.7 | glf > 0.7 |
| Model data                                                                      | glf < 0.6       | 5994            | 70        |
|                                                                                 | 0.6 < glf < 0.7 | 17              | 7760      |
|                                                                                 | glf > 0.7       | 48              | 26        |
| Number of points 22195 ; PC=99.0%; EBD=0.5%; POD1=98.9%; POD2=98.8%; POD3=98.6% |                 |                 |           |

Table 22: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

## 2 DAY1: Contingency tables

Tables in section 2 are included in the range: 23-32.

| wind speed                                                                      |                | Observations |                |          |
|---------------------------------------------------------------------------------|----------------|--------------|----------------|----------|
| $ms^{-1}$                                                                       |                | ws < 4.6     | 4.6 < ws < 7.4 | ws > 7.4 |
| Model data                                                                      | ws < 4.6       | 6201         | 1834           | 414      |
|                                                                                 | 4.6 < ws < 7.4 | 5200         | 7458           | 2630     |
|                                                                                 | ws > 7.4       | 481          | 2991           | 10013    |
| Number of points 37222 ; PC=63.6%; EBD=2.4%; POD1=52.2%; POD2=60.7%; POD3=76.7% |                |              |                |          |

Table 23: day1 - Contingency table for wind speed ( $ms^{-1}$ ) in standard configuration

| wind speed<br>$ms^{-1}$                                                         |                | Observations |                |          |
|---------------------------------------------------------------------------------|----------------|--------------|----------------|----------|
|                                                                                 |                | ws < 4.6     | 4.6 < ws < 7.4 | ws > 7.4 |
| Model data                                                                      | ws < 4.6       | 10592        | 896            | 8        |
|                                                                                 | 4.6 < ws < 7.4 | 1284         | 10710          | 629      |
|                                                                                 | ws > 7.4       | 6            | 677            | 12420    |
| Number of points 37222 ; PC=90.6%; EBD=0.0%; POD1=89.1%; POD2=87.2%; POD3=95.1% |                |              |                |          |

Table 24: day1 - Contingency table for wind speed ( $ms^{-1}$ ) processed with AR (1H)

| wind speed<br>$ms^{-1}$                                                         | Observations     |                  |           |     |
|---------------------------------------------------------------------------------|------------------|------------------|-----------|-----|
|                                                                                 | ws < 12.0        | 12.0 < ws < 18.0 | ws > 18.0 |     |
|                                                                                 | ws < 12.0        | 31937            | 401       | 0   |
| Model data                                                                      | 12.0 < ws < 18.0 | 294              | 3725      | 91  |
|                                                                                 | ws > 18.0        | 0                | 109       | 665 |
| Number of points 37222 ; PC=97.6%; EBD=0.0%; POD1=99.1%; POD2=88.0%; POD3=88.0% |                  |                  |           |     |

Table 25: day1 - Contingency table for wind speed ( $ms^{-1}$ ) processed with AR (1H) and thresholds (12.0,18.0)

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| wind direction<br>degree                                                                    |       | North | East | South | Observations | West |
|---------------------------------------------------------------------------------------------|-------|-------|------|-------|--------------|------|
| Model data                                                                                  | North | 13510 | 1129 | 10    |              | 282  |
|                                                                                             | East  | 263   | 859  | 138   |              | 1    |
|                                                                                             | South | 62    | 65   | 2091  |              | 798  |
|                                                                                             | West  | 4934  | 43   | 306   |              | 8089 |
| Number of points 32580 ; PC=75.3%; EBD=0.4%; POD1=72.0%; POD2=41.0%; POD3=82.2%; POD4=88.2% |       |       |      |       |              |      |

Table 26: day1 - Contingency table for wind direction (degree) in standard configuration

| wind direction<br>degree                                                                    |       | North | East | South | Observations | West |
|---------------------------------------------------------------------------------------------|-------|-------|------|-------|--------------|------|
| Model data                                                                                  | North | 18045 | 158  | 2     |              | 426  |
|                                                                                             | East  | 160   | 1908 | 14    |              | 0    |
|                                                                                             | South | 0     | 28   | 2405  |              | 107  |
|                                                                                             | West  | 564   | 2    | 124   |              | 8637 |
| Number of points 32580 ; PC=95.1%; EBD=0.0%; POD1=96.1%; POD2=91.0%; POD3=94.5%; POD4=94.2% |       |       |      |       |              |      |

Table 27: day1 - Contingency table for wind direction (degree) processed with AR (1H)

| relative humidity<br>percent                                                    |                 | Observations |                 |           |
|---------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                                 |                 | rh < 8.9     | 8.9 < rh < 17.9 | rh > 17.9 |
| Model data                                                                      | rh < 8.9        | 9856         | 2150            | 63        |
|                                                                                 | 8.9 < rh < 17.9 | 4996         | 8575            | 1926      |
|                                                                                 | rh > 17.9       | 329          | 2791            | 6488      |
| Number of points 37174 ; PC=67.0%; EBD=1.1%; POD1=64.9%; POD2=63.4%; POD3=76.5% |                 |              |                 |           |

Table 28: day1 - Contingency table for relative humidity (percent) in standard configuration

| relative humidity<br>percent                                                    |                 | Observations |                 |           |
|---------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                                 |                 | rh < 8.9     | 8.9 < rh < 17.9 | rh > 17.9 |
| Model data                                                                      | rh < 8.9        | 14478        | 566             | 1         |
|                                                                                 | 8.9 < rh < 17.9 | 703          | 12568           | 282       |
|                                                                                 | rh > 17.9       | 0            | 382             | 8194      |
| Number of points 37174 ; PC=94.8%; EBD=0.0%; POD1=95.4%; POD2=93.0%; POD3=96.7% |                 |              |                 |           |

Table 29: day1 - Contingency table for relative humidity (percent) processed with AR (1H)

| relative humidity<br>percent                                                    |                  | Observations |                  |           |
|---------------------------------------------------------------------------------|------------------|--------------|------------------|-----------|
|                                                                                 |                  | rh < 50.0    | 50.0 < rh < 70.0 | rh > 70.0 |
| Model data                                                                      | rh < 50.0        | 36357        | 36               | 0         |
|                                                                                 | 50.0 < rh < 70.0 | 38           | 652              | 10        |
|                                                                                 | rh > 70.0        | 0            | 2                | 79        |
| Number of points 37174 ; PC=99.8%; EBD=0.0%; POD1=99.9%; POD2=94.5%; POD3=88.8% |                  |              |                  |           |

Table 30: day1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

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| precipitable water vapor<br>mm                                                  | Observations    |           |                 |           |
|---------------------------------------------------------------------------------|-----------------|-----------|-----------------|-----------|
|                                                                                 |                 | pwv < 2.1 | 2.1 < pwv < 3.7 | pwv > 3.7 |
| Model data                                                                      | pwv < 2.1       | 10641     | 1494            | 0         |
|                                                                                 | 2.1 < pwv < 3.7 | 2093      | 5257            | 902       |
|                                                                                 | pwv > 3.7       | 93        | 1117            | 6444      |
| Number of points 28041 ; PC=79.7%; EBD=0.3%; POD1=83.0%; POD2=66.8%; POD3=87.7% |                 |           |                 |           |

Table 31: day1 - Contingency table for precipitable water vapor (mm) in standard configuration

| precipitable water vapor<br>mm                                                  | Observations    |           |                 |           |
|---------------------------------------------------------------------------------|-----------------|-----------|-----------------|-----------|
|                                                                                 |                 | pwv < 2.1 | 2.1 < pwv < 3.7 | pwv > 3.7 |
| Model data                                                                      | pwv < 2.1       | 12557     | 143             | 0         |
|                                                                                 | 2.1 < pwv < 3.7 | 268       | 7593            | 121       |
|                                                                                 | pwv > 3.7       | 2         | 132             | 7225      |
| Number of points 28041 ; PC=97.6%; EBD=0.0%; POD1=97.9%; POD2=96.5%; POD3=98.4% |                 |           |                 |           |

Table 32: day1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

3 NIGHT2: Contingency tables

Tables in section 3 are included in the range: 33-42.

| wind speed<br>$ms^{-1}$                                                         |                | Observations |                |          |
|---------------------------------------------------------------------------------|----------------|--------------|----------------|----------|
|                                                                                 |                | ws < 4.2     | 4.2 < ws < 7.6 | ws > 7.6 |
| Model data                                                                      | ws < 4.2       | 7761         | 3054           | 368      |
|                                                                                 | 4.2 < ws < 7.6 | 4385         | 5819           | 2251     |
|                                                                                 | ws > 7.6       | 636          | 2369           | 9398     |
| Number of points 36041 ; PC=63.8%; EBD=2.8%; POD1=60.7%; POD2=51.8%; POD3=78.2% |                |              |                |          |

Table 33: night2 - Contingency table for wind speed ( $ms^{-1}$ ) in standard configuration

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| wind direction<br>degree | Observations |       |       |      |
|--------------------------|--------------|-------|-------|------|
|                          | North        | East  | South | West |
| Model data               | North        | 16013 | 990   | 115  |
|                          | East         | 1328  | 2489  | 505  |
|                          | South        | 130   | 581   | 3796 |
|                          | West         | 541   | 139   | 123  |

Number of points 27679 ; PC=81.8%; EBD=1.4%; POD1=88.9%; POD2=59.3%; POD3=83.6%; POD4=36.0%

Table 34: night2 - Contingency table for wind direction (degree) in standard configuration

| relative humidity<br>percent | Observations    |                 |           |
|------------------------------|-----------------|-----------------|-----------|
|                              | rh < 7.1        | 7.1 < rh < 15.3 | rh > 15.3 |
| Model data                   | rh < 7.1        | 8407            | 2525      |
|                              | 7.1 < rh < 15.3 | 5653            | 7227      |
|                              | rh > 15.3       | 697             | 2953      |

Number of points 35654 ; PC=62.8%; EBD=2.1%; POD1=57.0%; POD2=56.9%; POD3=82.3%

Table 35: night2 - Contingency table for relative humidity (percent) in standard configuration

| precipitable water vapor<br>mm | Observations    |                 |           |
|--------------------------------|-----------------|-----------------|-----------|
|                                | pwv < 1.9       | 1.9 < pwv < 3.5 | pwv > 3.5 |
| Model data                     | pwv < 1.9       | 10420           | 2231      |
|                                | 1.9 < pwv < 3.5 | 1837            | 4486      |
|                                | pwv > 3.5       | 84              | 1077      |

Number of points 27649 ; PC=76.0%; EBD=0.5%; POD1=84.4%; POD2=57.6%; POD3=81.2%

Table 36: night2 - Contingency table for precipitable water vapor (mm) in standard configuration

| total seeing     |                 | Observations |                                       |           |
|------------------|-----------------|--------------|---------------------------------------|-----------|
| arcsec           |                 | see < 0.6    | 0.6 < see < 0.8                       | see > 0.8 |
| Model data       | see < 0.6       | 4931         | 3991                                  | 1432      |
|                  | 0.6 < see < 0.8 | 3084         | 3833                                  | 3286      |
|                  | see > 0.8       | 626          | 1339                                  | 3966      |
| Number of points |                 | 26488        | ; PC=%; EBD=%; POD1=%; POD2=%; POD3=% |           |

Table 37: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

| total seeing     |                 | Observations |                                       |           |
|------------------|-----------------|--------------|---------------------------------------|-----------|
| arcsec           |                 | see < 0.6    | 0.6 < see < 0.8                       | see > 0.8 |
| Model data       | see < 0.6       | 8170         | 3                                     | 1034      |
|                  | 0.6 < see < 0.8 | 0            | 8869                                  | 1029      |
|                  | see > 0.8       | 471          | 291                                   | 7650      |
| Number of points |                 | 26488        | ; PC=%; EBD=%; POD1=%; POD2=%; POD3=% |           |

Table 38: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

| wavefront coherence time                                                        |                    | Observations |                    |              |
|---------------------------------------------------------------------------------|--------------------|--------------|--------------------|--------------|
| ms                                                                              |                    | $\tau < 3.4$ | $3.4 < \tau < 5.8$ | $\tau > 5.8$ |
| Model data                                                                      | $\tau < 3.4$       | 4749         | 2854               | 314          |
|                                                                                 | $3.4 < \tau < 5.8$ | 1628         | 4648               | 2977         |
|                                                                                 | $\tau > 5.8$       | 261          | 2121               | 5697         |
| Number of points 25249 ; PC=59.8%; EBD=2.3%; POD1=71.5%; POD2=48.3%; POD3=63.4% |                    |              |                    |              |

Table 39: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

| wavefront coherence time                                                        |                    | Observations |                    |              |
|---------------------------------------------------------------------------------|--------------------|--------------|--------------------|--------------|
| ms                                                                              |                    | $\tau < 3.4$ | $3.4 < \tau < 5.8$ | $\tau > 5.8$ |
| Model data                                                                      | $\tau < 3.4$       | 5942         | 375                | 1308         |
|                                                                                 | $3.4 < \tau < 5.8$ | 435          | 8141               | 1308         |
|                                                                                 | $\tau > 5.8$       | 261          | 1107               | 7680         |
| Number of points 25249 ; PC=86.2%; EBD=6.2%; POD1=89.5%; POD2=84.6%; POD3=74.6% |                    |              |                    |              |

Table 40: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

| ground layer fraction<br>dimensionless                                           | Observations    |                 |           |
|----------------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                                  | glf < 0.6       | 0.6 < glf < 0.7 | glf > 0.7 |
| Model data                                                                       | glf < 0.6       | 3865            | 4326      |
|                                                                                  | 0.6 < glf < 0.7 | 2616            | 3629      |
|                                                                                  | glf > 0.7       | 626             | 1085      |
| Number of points 25626 ; PC=37.5%; EBD=15.4%; POD1=54.4%; POD2=40.1%; POD3=22.4% |                 |                 |           |

Table 41: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

| ground layer fraction<br>dimensionless                                           | Observations    |                 |           |
|----------------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                                  | glf < 0.6       | 0.6 < glf < 0.7 | glf > 0.7 |
| Model data                                                                       | glf < 0.6       | 6371            | 1309      |
|                                                                                  | 0.6 < glf < 0.7 | 110             | 7683      |
|                                                                                  | glf > 0.7       | 626             | 48        |
| Number of points 25626 ; PC=77.8%; EBD=16.5%; POD1=89.6%; POD2=85.0%; POD3=47.7% |                 |                 |           |

Table 42: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

4 DAY2: Contingency tables

Tables in section 4 are included in the range: 43-46.

| wind speed<br>$ms^{-1}$                                                         |                | Observations |                |          |
|---------------------------------------------------------------------------------|----------------|--------------|----------------|----------|
|                                                                                 |                | ws < 4.6     | 4.6 < ws < 7.4 | ws > 7.4 |
| Model data                                                                      | ws < 4.6       | 6062         | 1780           | 459      |
|                                                                                 | 4.6 < ws < 7.4 | 5337         | 7294           | 2610     |
|                                                                                 | ws > 7.4       | 425          | 2956           | 9519     |
| Number of points 36442 ; PC=62.8%; EBD=2.4%; POD1=51.3%; POD2=60.6%; POD3=75.6% |                |              |                |          |

Table 43: day2 - Contingency table for wind speed ( $ms^{-1}$ ) in standard configuration

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| wind direction<br>degree | Observations |       |       |      |
|--------------------------|--------------|-------|-------|------|
|                          | North        | East  | South | West |
| Model data               | North        | 13015 | 1120  | 35   |
|                          | East         | 283   | 807   | 138  |
|                          | South        | 83    | 98    | 2022 |
|                          | West         | 4766  | 44    | 402  |

Number of points 31696 ; PC=74.8%; EBD=0.5%; POD1=71.7%; POD2=39.0%; POD3=77.9%; POD4=88.6%

Table 44: day2 - Contingency table for wind direction (degree) in standard configuration

| relative humidity<br>percent | Observations    |                 |           |
|------------------------------|-----------------|-----------------|-----------|
|                              | rh < 8.9        | 8.9 < rh < 17.9 | rh > 17.9 |
| Model data                   | rh < 8.9        | 9431            | 1984      |
|                              | 8.9 < rh < 17.9 | 5533            | 8057      |
|                              | rh > 17.9       | 386             | 2610      |

Number of points 36373 ; PC=65.5%; EBD=1.3%; POD1=61.4%; POD2=63.7%; POD3=75.8%

Table 45: day2 - Contingency table for relative humidity (percent) in standard configuration

| precipitable water vapor<br>mm | Observations    |                 |           |
|--------------------------------|-----------------|-----------------|-----------|
|                                | pwv < 2.1       | 2.1 < pwv < 3.7 | pwv > 3.7 |
| Model data                     | pwv < 2.1       | 10834           | 1676      |
|                                | 2.1 < pwv < 3.7 | 2287            | 4716      |
|                                | pwv > 3.7       | 102             | 1134      |

Number of points 28365 ; PC=77.1%; EBD=0.4%; POD1=81.9%; POD2=62.7%; POD3=82.8%

Table 46: day2 - Contingency table for precipitable water vapor (mm) in standard configuration

5 NIGHT3: Contingency tables

Tables in section 5 are included in the range: 47-56.

| wind speed<br>$ms^{-1}$                                                         |                | Observations |                |          |
|---------------------------------------------------------------------------------|----------------|--------------|----------------|----------|
|                                                                                 |                | ws < 4.2     | 4.2 < ws < 7.6 | ws > 7.6 |
| Model data                                                                      | ws < 4.2       | 7101         | 3313           | 501      |
|                                                                                 | 4.2 < ws < 7.6 | 4973         | 5464           | 2583     |
|                                                                                 | ws > 7.6       | 711          | 2613           | 8849     |
| Number of points 36108 ; PC=59.3%; EBD=3.4%; POD1=55.5%; POD2=48.0%; POD3=74.2% |                |              |                |          |

Table 47: night3 - Contingency table for wind speed ( $ms^{-1}$ ) in standard configuration

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| wind direction<br>degree | Observations |       |       |      |
|--------------------------|--------------|-------|-------|------|
|                          | North        | East  | South | West |
| Model data               | North        | 15551 | 1029  | 114  |
|                          | East         | 1647  | 2461  | 662  |
|                          | South        | 281   | 668   | 3631 |
|                          | West         | 587   | 124   | 97   |

Number of points 27773 ; PC=78.8%; EBD=1.9%; POD1=86.1%; POD2=57.5%; POD3=80.6%; POD4=27.7%

Table 48: night3 - Contingency table for wind direction (degree) in standard configuration

| relative humidity<br>percent | Observations    |                 |           |
|------------------------------|-----------------|-----------------|-----------|
|                              | rh < 7.1        | 7.1 < rh < 15.3 | rh > 15.3 |
| Model data                   | rh < 7.1        | 7997            | 2721      |
|                              | 7.1 < rh < 15.3 | 6016            | 6872      |
|                              | rh > 15.3       | 751             | 3007      |

Number of points 35654 ; PC=59.9%; EBD=2.7%; POD1=54.2%; POD2=54.5%; POD3=78.2%

Table 49: night3 - Contingency table for relative humidity (percent) in standard configuration

| precipitable water vapor<br>mm | Observations    |                 |           |
|--------------------------------|-----------------|-----------------|-----------|
|                                | pwv < 1.9       | 1.9 < pwv < 3.5 | pwv > 3.5 |
| Model data                     | pwv < 1.9       | 10479           | 2461      |
|                                | 1.9 < pwv < 3.5 | 1583            | 4163      |
|                                | pwv > 3.5       | 223             | 1168      |

Number of points 27464 ; PC=73.8%; EBD=1.5%; POD1=85.3%; POD2=53.4%; POD3=76.1%

Table 50: night3 - Contingency table for precipitable water vapor (mm) in standard configuration

| total seeing     |                 | Observations |                                       |           |
|------------------|-----------------|--------------|---------------------------------------|-----------|
| arcsec           |                 | see < 0.6    | 0.6 < see < 0.8                       | see > 0.8 |
| Model data       | see < 0.6       | 4808         | 3960                                  | 1571      |
|                  | 0.6 < see < 0.8 | 2969         | 3704                                  | 3237      |
|                  | see > 0.8       | 812          | 1530                                  | 3851      |
| Number of points |                 | 26442        | ; PC=%; EBD=%; POD1=%; POD2=%; POD3=% |           |

Table 51: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

| total seeing     |                 | Observations |                                       |           |
|------------------|-----------------|--------------|---------------------------------------|-----------|
| arcsec           |                 | see < 0.6    | 0.6 < see < 0.8                       | see > 0.8 |
| Model data       | see < 0.6       | 7964         | 0                                     | 1083      |
|                  | 0.6 < see < 0.8 | 0            | 8887                                  | 1083      |
|                  | see > 0.8       | 625          | 307                                   | 7576      |
| Number of points |                 | 26442        | ; PC=%; EBD=%; POD1=%; POD2=%; POD3=% |           |

Table 52: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

| wavefront coherence time                                                        |                    | Observations |                    |              |
|---------------------------------------------------------------------------------|--------------------|--------------|--------------------|--------------|
| ms                                                                              |                    | $\tau < 3.4$ | $3.4 < \tau < 5.8$ | $\tau > 5.8$ |
| Model data                                                                      | $\tau < 3.4$       | 4525         | 2832               | 326          |
|                                                                                 | $3.4 < \tau < 5.8$ | 1733         | 4400               | 2778         |
|                                                                                 | $\tau > 5.8$       | 390          | 2450               | 5860         |
| Number of points 25294 ; PC=58.5%; EBD=2.8%; POD1=68.1%; POD2=45.4%; POD3=65.4% |                    |              |                    |              |

Table 53: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

| wavefront coherence time                                                        |                    | Observations |                    |              |
|---------------------------------------------------------------------------------|--------------------|--------------|--------------------|--------------|
| ms                                                                              |                    | $\tau < 3.4$ | $3.4 < \tau < 5.8$ | $\tau > 5.8$ |
| Model data                                                                      | $\tau < 3.4$       | 5760         | 322                | 1352         |
|                                                                                 | $3.4 < \tau < 5.8$ | 498          | 8056               | 1352         |
|                                                                                 | $\tau > 5.8$       | 390          | 1304               | 7612         |
| Number of points 25294 ; PC=84.7%; EBD=6.9%; POD1=86.6%; POD2=83.2%; POD3=73.8% |                    |              |                    |              |

Table 54: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

| ground layer fraction<br>dimensionless                                           |                 | Observations |                 |           |
|----------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                                  |                 | glf < 0.6    | 0.6 < glf < 0.7 | glf > 0.7 |
| Model data                                                                       | glf < 0.6       | 3529         | 3948            | 3213      |
|                                                                                  | 0.6 < glf < 0.7 | 2689         | 3944            | 4079      |
|                                                                                  | glf > 0.7       | 835          | 1135            | 2236      |
| Number of points 25608 ; PC=37.9%; EBD=15.8%; POD1=50.0%; POD2=43.7%; POD3=23.5% |                 |              |                 |           |

Table 55: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

| ground layer fraction<br>dimensionless                                           |                 | Observations |                 |           |
|----------------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                                  |                 | glf < 0.6    | 0.6 < glf < 0.7 | glf > 0.7 |
| Model data                                                                       | glf < 0.6       | 6112         | 1119            | 3477      |
|                                                                                  | 0.6 < glf < 0.7 | 106          | 7872            | 2698      |
|                                                                                  | glf > 0.7       | 835          | 36              | 6051      |
| Number of points 25608 ; PC=78.2%; EBD=16.8%; POD1=86.7%; POD2=87.2%; POD3=49.5% |                 |              |                 |           |

Table 56: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

6 DAY3: Contingency tables

Tables in section 6 are included in the range: 57-60.

| wind speed<br>$ms^{-1}$                                                         |                | Observations |                |          |
|---------------------------------------------------------------------------------|----------------|--------------|----------------|----------|
|                                                                                 |                | ws < 4.6     | 4.6 < ws < 7.4 | ws > 7.4 |
| Model data                                                                      | ws < 4.6       | 5849         | 1888           | 454      |
|                                                                                 | 4.6 < ws < 7.4 | 5460         | 7260           | 2719     |
|                                                                                 | ws > 7.4       | 450          | 2887           | 9408     |
| Number of points 36375 ; PC=61.9%; EBD=2.5%; POD1=49.7%; POD2=60.3%; POD3=74.8% |                |              |                |          |

Table 57: day3 - Contingency table for wind speed ( $ms^{-1}$ ) in standard configuration

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| wind direction<br>degree | Observations |       |       |      |
|--------------------------|--------------|-------|-------|------|
|                          | North        | East  | South | West |
| Model data               | North        | 12683 | 1149  | 79   |
|                          | East         | 303   | 771   | 124  |
|                          | South        | 135   | 78    | 1904 |
|                          | West         | 5018  | 65    | 489  |

Number of points 31668 ; PC=72.6%; EBD=0.9%; POD1=69.9%; POD2=37.4%; POD3=73.3%; POD4=85.9%

Table 58: day3 - Contingency table for wind direction (degree) in standard configuration

| relative humidity<br>percent | Observations    |                 |           |
|------------------------------|-----------------|-----------------|-----------|
|                              | rh < 8.9        | 8.9 < rh < 17.9 | rh > 17.9 |
| Model data                   | rh < 8.9        | 8867            | 1951      |
|                              | 8.9 < rh < 17.9 | 5983            | 7883      |
|                              | rh > 17.9       | 488             | 2792      |

Number of points 36306 ; PC=62.9%; EBD=1.7%; POD1=57.8%; POD2=62.4%; POD3=73.0%

Table 59: day3 - Contingency table for relative humidity (percent) in standard configuration

| precipitable water vapor<br>mm | Observations    |                 |           |
|--------------------------------|-----------------|-----------------|-----------|
|                                | pwv < 2.1       | 2.1 < pwv < 3.7 | pwv > 3.7 |
| Model data                     | pwv < 2.1       | 10685           | 2046      |
|                                | 2.1 < pwv < 3.7 | 2385            | 4233      |
|                                | pwv > 3.7       | 132             | 1201      |

Number of points 28286 ; PC=74.3%; EBD=0.8%; POD1=80.9%; POD2=56.6%; POD3=80.4%

Table 60: day3 - Contingency table for precipitable water vapor (mm) in standard configuration